

MEDTRACK DV

HOSPITAL PERFORMANCE ANALYTICS DASHBOARD

Quality Assurance Checklist

Pre-Delivery Functional & KPI Validation Report

Project Name	MedTrack DV – Hospital Performance Analytics Dashboard
Prepared By	Madhan Raj G
Role	Data Visualization Intern
Tool Used	Tableau Public
Document Version	1.0
Document Status	Final
Testing Date	_____
Classification	Internal – Infosys Springboard Submission

Infosys Springboard Internship — Data Visualization Track

Document Control

This section records the revision history and ownership of this QA checklist for traceability.

Revision History

Version	Date	Author	Description of Change
0.1	10-Jul-2026	Madhan Raj G	Initial draft – checklist structure and scope defined
0.2	16-Jul-2026	Madhan Raj G	Functional and KPI test cases added after Module 5 build
1.0	22-Jul-2026	Madhan Raj G	Final review completed; document approved for submission

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1. Executive Summary

This report summarizes the Quality Assurance activity carried out on the MedTrack DV Hospital Performance Analytics Dashboard, developed on Tableau Public as part of the Infosys Springboard Data Visualization internship. Testing was carried out across all four dashboards covering functional behavior, KPI accuracy, navigation, filters, and visual consistency. The results below reflect the outcome of that testing cycle.

26 Test Cases Executed	26 Passed	0 Failed	100% Pass Rate	19 KPIs Validated
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2. Document Purpose

This Quality Assurance (QA) checklist has been prepared to validate the functionality, accuracy, and overall quality of the MedTrack DV Hospital Performance Analytics Dashboard prior to final delivery and submission. The purpose of this document is to systematically verify dashboard functionality, KPI calculations, filter behavior, navigation controls, and visualization accuracy, ensuring that the deliverable meets the standard expected of a production-grade business intelligence solution. This checklist follows conventional software QA practice adapted for Tableau-based analytics dashboards and stands as evidence that the project underwent structured testing before being marked complete.

3. Scope

The scope of this testing effort covers the following areas of the MedTrack DV dashboard suite:

- Dashboard functionality across all four dashboards
- KPI validation and calculation accuracy
- Chart and visualization validation
- Navigation buttons and inter-dashboard linking
- Interactive filters and their downstream effects
- Dashboard formatting, layout, and visual consistency
- Overall user experience (UX) and dashboard responsiveness

Testing was restricted to the published Tableau Public workbook and the underlying hospital operations dataset; back-end data pipeline testing was outside the scope of this exercise.

4. Testing Methodology

Testing was performed manually through direct interaction with the published dashboard, using a combination of functional walkthroughs and independent recalculation of KPI values against the source dataset.

- Functional testing: each interactive element (filter, button, tooltip) was exercised manually and the resulting behavior compared against the expected outcome.
- KPI validation: every headline metric was recalculated outside Tableau using the raw dataset and cross-checked against the value shown on the dashboard.
- Visual review: each dashboard was inspected at standard desktop resolution for layout, alignment, and formatting consistency.
- Regression pass: a final full pass was completed after all Module 6 changes were merged, prior to sign-off.

5. Test Environment

Field	Details
Application	MedTrack DV – Hospital Performance Analytics Dashboard
Tool Used	Tableau Public
Dataset	Hospital Operations Dataset (2,238 records / 37 fields)
Operating System	Windows 11 (64-bit)
Browser	Google Chrome (latest stable release)
Testing Type	Functional Testing, UI/UX Testing, KPI Validation Testing

6. Functional QA Checklist

The following functional test cases were executed across all four dashboards within the MedTrack DV project. Each case was tested by Madhan Raj G on the final published build.

ID	Module	Test Case	Expected Result	Actual Result	Status
TC-001	Overview	Dashboard loads successfully	Dashboard opens without errors	Dashboard loaded successfully	PASS
TC-002	Overview	Hospital Overview Dashboard loads	All visuals render correctly	All visuals rendered correctly	PASS
TC-003	Patient Flow	Patient Flow Dashboard loads	All visuals render correctly	All visuals rendered correctly	PASS

ID	Module	Test Case	Expected Result	Actual Result	Status
TC-004	Dept. Analytics	Department Analytics Dashboard loads	All visuals render correctly	All visuals rendered correctly	PASS
TC-005	Resource Util.	Resource Utilization Dashboard loads	All visuals render correctly	All visuals rendered correctly	PASS
TC-006	Filters	Filters update charts correctly	Charts refresh instantly on filter selection	Charts refreshed as expected	PASS
TC-007	Navigation	Navigation buttons work	User is redirected to correct dashboard	Redirection worked correctly	PASS
TC-008	Navigation	Home button works	Returns user to Hospital Overview Dashboard	Returned to Overview correctly	PASS
TC-009	Navigation	Dashboard button works	Navigates to selected dashboard	Navigation successful	PASS
TC-010	Navigation	About button works	Displays project/about information	About panel displayed correctly	PASS
TC-011	KPI	KPI values display correctly	KPI cards show accurate calculated values	KPI values matched dataset	PASS

ID	Module	Test Case	Expected Result	Actual Result	Status
TC-012	Filters	Charts update after filters	All linked charts reflect filter changes	All linked charts updated	PASS
TC-013	UI	Tooltips display properly	Tooltips show correct data on hover	Tooltips displayed correct data	PASS
TC-014	UI	Titles are correct	Dashboard and chart titles match content	Titles verified as accurate	PASS
TC-015	UI	Formatting is consistent	Fonts, colors, and spacing consistent across dashboards	Formatting consistent throughout	PASS
TC-016	Performance	Dashboard responsiveness	Dashboard resizes properly across screen sizes	Responsive behavior confirmed	PASS
TC-017	UI	No overlapping objects	All charts and objects are properly aligned	No overlapping objects found	PASS
TC-018	UI	No missing charts	All expected charts are present on each dashboard	All charts present and accounted for	PASS

ID	Module	Test Case	Expected Result	Actual Result	Status
TC-019	KPI	No broken calculations	All calculated fields return valid results	No broken calculations found	PASS
TC-020	Performance	Performance is smooth	Dashboard loads and interacts without lag	Performance smooth, no lag observed	PASS
TC-021	Filters	Filter reset works correctly	All filters reset to default state on clear	Filters reset successfully	PASS
TC-022	UI	Color legend accuracy	Legend colors match chart data series	Legend colors verified accurate	PASS
TC-023	Filters	Hospital Overview filters (Month of Admission, Admission Type, Hospital Region, Department) update all linked visuals	All 9 charts and 6 KPI cards refresh on selection	Verified across all four filter controls	PASS
TC-024	Filters	Patient Flow filters (Age Group, Severity, Discharge Status) update all linked visuals	Severity Distribution, Discharge Status, and Age Group charts refresh correctly	Verified across all three filter controls	PASS

ID	Module	Test Case	Expected Result	Actual Result	Status
TC-025	Filters	Department Analytics filters (Test Results, Medical Condition, Admission Type) update all linked visuals	Department-level charts and heat-map refresh on selection	Verified across all filter controls	PASS
TC-026	Filters	Resource Utilization filters (Equipment Used, Room Type, Ward, Staff Shift) update all linked visuals	Bed, equipment, and staffing charts refresh on selection	Verified across all four filter controls	PASS

7. KPI Validation Checklist

Every Key Performance Indicator shown across all four dashboards was independently recalculated and cross-verified against the source dataset to confirm accuracy. KPIs are grouped below by the dashboard on which they appear.

7.1 Hospital Overview Dashboard

KPI	Validation Method	Status
Total Patients	Dashboard shows 2,238 – cross-checked against COUNT of patient records in source dataset (2,238 rows)	Validated
Total Billing	Dashboard shows \$56,610,270 – cross-checked against SUM of billing amount field	Validated
Bed Occupancy Rate	Dashboard shows 80.8% – recalculated as 1,809 occupied / 2,238 total beds = 80.83%	Validated
Avg Length of Stay	Dashboard shows 15.5 days – recalculated as AVG of admission-to-discharge duration across all admission types	Validated
Readmission Rate	Dashboard shows 9.1% – recalculated as 204 readmitted / 2,238 total patients = 9.12% (Readmission Analysis chart)	Validated
Discharge Rate	Dashboard shows 34.1% – verified against discharge status field proportions	Validated

7.2 Patient Flow Dashboard

KPI	Validation Method	Status
Total Admissions	Dashboard shows 2,238 – matches Total Patients on Hospital Overview Dashboard (cross-dashboard consistency check)	Validated
Occupied Beds	Dashboard shows 1,809 – matches Bed Occupancy Status chart (Occupied segment = 1,809)	Validated
Severe Cases %	Dashboard shows 23.9% – recalculated as 536 severe cases / 2,238 total = 23.95% (Severity Distribution chart)	Validated
ICU Admissions	Dashboard shows 339 – matches sum of ICU Admissions by Ward on Resource Utilization Dashboard (81+69+67+61+61 = 339)	Validated
Emergency Cases	Dashboard shows 472 – matches sum of 'Yes' values in Emergency Cases by Department on Department Analytics Dashboard	Validated
Average Billing per Patient	Dashboard shows \$25,295 – recalculated as Total Billing / Total Patients (\$56,610,270 / 2,238 = \$25,295)	Validated

7.3 Department Analytics Dashboard

KPI	Validation Method	Status
Total Departments	Dashboard shows 6 – verified against distinct COUNT of department field (Cardiology, Endocrinology, Nutrition, Oncology, Orthopedics, Pulmonology)	Validated
Avg Patients per Department	Dashboard shows 373.0 – recalculated as Total Patients / Total Departments (2,238 / 6 = 373.0)	Validated
Avg Billing per Department	Dashboard shows \$9,435,045 – recalculated as Total Billing / Total Departments (\$56,610,270 / 6 = \$9,435,045)	Validated

7.4 Resource Utilization Dashboard

KPI	Validation Method	Status
Total Nurses	Dashboard shows 50 – cross-checked against staffing dataset field	Validated
Available Beds	Dashboard shows 429 – matches Bed Occupancy Status 'Vacant' segment on Hospital Overview and Patient Flow Dashboards	Validated
Avg Patients per Doctor	Dashboard shows 1.015 – recalculated as Total Patients / Total Doctors on record	Validated
Total Wards	Dashboard shows 5 – verified against distinct COUNT of ward field (General, Semi-Private, Emergency, Private, ICU)	Validated

8. Dashboard Validation Summary

Dashboard	Filters Verified	Tested By	Result
Hospital Overview	Month of Admission, Admission Type, Hospital Region, Department	Madhan Raj G	PASS
Patient Flow	Month of Admission, Age Group, Severity, Discharge Status	Madhan Raj G	PASS
Department Analytics	Test Results, Medical Condition, Admission Type, Department	Madhan Raj G	PASS
Resource Utilization	Equipment Used, Room Type, Ward, Staff Shift	Madhan Raj G	PASS

9. Issue Log

No major issues were identified during testing. All dashboards, KPIs, navigation controls, filters, and visualizations functioned as expected.

Issue ID	Description	Severity	Status
—	No defects were logged during this testing cycle.	N/A	N/A

10. Recommendations

- Continue monitoring KPI calculations whenever the underlying dataset is updated or refreshed.
- Perform regression testing before publishing any future releases or dashboard revisions.
- Maintain consistent formatting, color palette, and typography standards across all dashboards.
- Validate any newly introduced calculated fields thoroughly before deployment.
- Periodically review filter and navigation performance as the dataset scales in size.

11. Final QA Summary

All dashboard functionality, KPI calculations, navigation controls, filters, and visualizations were successfully validated through structured functional and KPI-level testing.

The MedTrack DV Tableau project has passed Quality Assurance testing and is confirmed ready for deployment and final project submission under the Infosys Springboard Internship program.